Facial Emotion Recognition, Levelled Up: Vision Transformer, Attention Explanations and Evaluation under Talking-Head Animation

Computer Vision — Assignment 2 (Continuation of Assignment 1)

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Abstract. We upgraded our Assignment 1 project — recognising *happy*, *sad* and *neutral* faces in a self-collected dataset — with state-of-the-art components. The dataset was expanded to 1,151 images and re-split so that every photo and all of its edited variants stay in the same partition, which removed a train/test leak present in Assignment 1. On this clean split, a fine-tuned Vision Transformer (ViT-B/16) reaches 0.884 accuracy and 0.891 macro-F1 on 172 test images, compared with 0.659 macro-F1 for the best Assignment 1 baseline (HOG + SVM). Attention rollout shows that the ViT focuses on the mouth, eyes and nose. Following the VASA-1 concept, we animated unseen test portraits into talking-head clips with LivePortrait: under head motion alone the predictions stay stable (Δmacro-F1 = +0.01), but when the mouth moves as in speech, macro-F1 falls from 0.870 to 0.483 (Δ = −0.39), revealing a strong reliance on mouth shape. A Gradio app provides upload/webcam prediction with an attention heatmap and latency.

1. Introduction

In Assignment 1 we collected face photographs of three emotions and compared HOG + SVM, a small CNN and MobileNetV2. Assignment 2 asks us to keep the topic, upgrade it with SOTA computer-vision components, explain the predictions, reflect on accuracy, latency and ethics, and deliver a working demo. For facial emotion recognition, the suggested SOTA add-on is **evaluation under animation**: turning a still portrait into a talking-head video, as in VASA-1 [3, 4], and checking whether the emotion prediction stays consistent frame by frame. Our contributions are: (1) a cleaned and expanded dataset with a leak-free, group-aware split; (2) a ViT-B/16 classifier [1] compared against the re-run A1 baselines, with an augmentation ablation; (3) attention-rollout explanations [2]; (4) a VASA-1-style animation test implemented with LivePortrait [5]; and (5) an interactive app reporting latency.

2. Dataset Update

The dataset keeps the Assignment 1 folder structure (one folder per emotion) and was expanded with new photographs taken by the group. Besides the original photographs, it contains **offline augmented variants** that we created ourselves: from one source photo up to 15 variants with changed brightness and contrast, mirroring, small rotations and soft shadows. These are classical image edits, not AI-generated images; the share of **AI-generated (synthetic) images in training is 0 %**. AI-generated media appear only in the animation test (Section 5), and are labelled as such.

Cleaning and leak-free split. While preparing Assignment 2 we found that the Assignment 1 split was made per image, so variants of the same photo could end up in both training and test, which inflated the A1 scores. The new pipeline therefore (i) removes byte-identical duplicates (3 files); (ii) groups every variant with its source photo using the file names (suffixes such as `_aug05`, `_aug_05`, `_original` and `(1)`); (iii) merges near-identical shots using a 64-bit difference hash that is also compared against the mirrored image; and (iv) assigns whole groups to train / validation / test so that each emotion receives about 70 / 15 / 15 % of its images (seed 42). Automatic checks confirm that no group, no identical file and no near-duplicate is shared between partitions. The result is 1,151 images from 159 source photographs and 992 offline variants, forming 149 independent photo groups (Table 1).

Table 1. Images per split and emotion; the last column gives independent photo groups (happy / sad / neutral). Training additionally uses on-the-fly augmentation.

Split	Happy	Sad	Neutral	Total	Photo groups
Train	253	188	365	806	20 / 14 / 32
Validation	54	41	78	173	17 / 12 / 13
Test	54	40	78	172	17 / 11 / 13
Total	**361**	**269**	**521**	**1,151**	**54 / 37 / 58**

The classes are imbalanced (neutral 45 %, sad 23 %), so all deep models use class-weighted cross-entropy. During training, images are additionally augmented on the fly with random resized crops (scale 0.85–1), horizontal flips, rotations of up to ±15° and colour jitter; validation and test images are never augmented.

3. SOTA Methods

3.1 Vision Transformer (ViT-B/16)

A Vision Transformer [1] treats an image like a sentence. The 224 × 224 face is cut into 196 patches of 16 × 16 pixels; each patch is flattened and linearly projected into a 768-dimensional token, a learned position embedding is added, and a special *[CLS]* token is prepended. Twelve Transformer encoder layers then apply multi-head self-attention, in which every patch can directly compare itself with every other patch, followed by an MLP. The final [CLS] token is classified by a linear head (Figure 1). **Why it helps our topic:** an expression is a relation between distant facial regions — mouth corners, eyes and eyebrows change together — and self-attention models such relations from the first layer, whereas a CNN builds them up slowly from local filters. Its attention weights also give a built-in explanation. We fine-tune the ImageNet-21k checkpoint `google/vit-base-patch16-224-in21k` end to end with AdamW (backbone lr $2{\cdot}10^{-5}$, new head 10^{-3}, weight decay 0.01), batch 16, up to 15 epochs with early stopping on validation loss (patience 5).

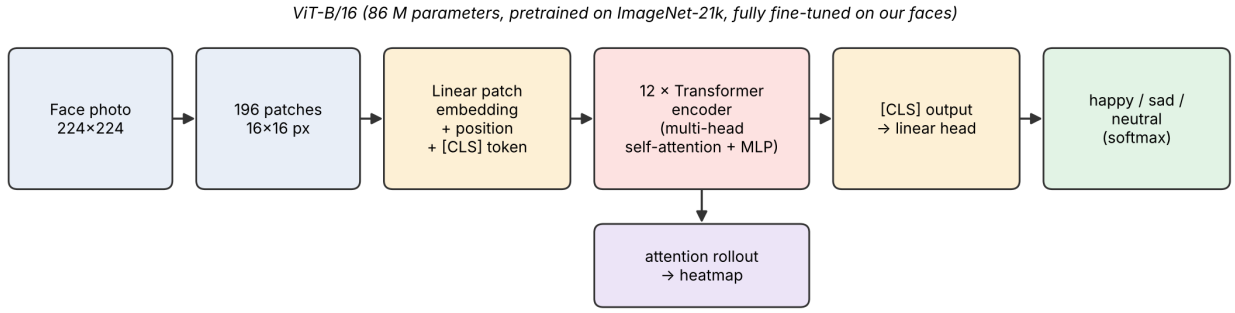


Figure 1. ViT-B/16 pipeline used in this work, with attention rollout for explanations.

3.2 VASA-1 concept: evaluation under talking-head animation

VASA-1 (Microsoft Research Asia, 2024) turns a single portrait and a speech clip into a lifelike talking-face video in real time, with synchronised lips, natural head motion and facial dynamics generated in a disentangled face latent space [3, 4]. Microsoft did not release the model because of deepfake concerns, so — as the assignment allows — we implement the concept with the open-source **LivePortrait** [5], which animates a portrait with the motion of a driving video (Figure 2). LivePortrait lets us restrict which facial region is animated: ***pose*** transfers only head motion, so the photographed expression is preserved; ***lip*** transfers only mouth motion, imitating speech. **Why it helps our topic:** a deployed emotion recogniser sees moving, talking people rather than posed stills; animating unseen test photos lets us measure, without collecting new video, whether predictions are stable over time and which motions break them. We animated 8 test photos per emotion with one 6-second talking reference video in both modes (48 clips) and evaluated every 3rd frame (624 frames per mode).

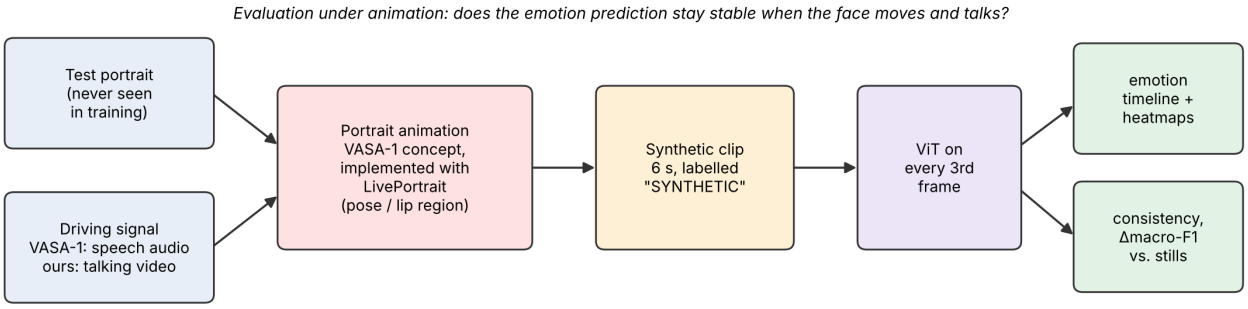


Figure 2. Evaluation under animation (VASA-1 concept implemented with LivePortrait).

The assignment also lists **ArtAug** [9] for improving difficult cases. ArtAug is a LoRA enhancement for the 12-billion-parameter FLUX.1-dev text-to-image model and could not be run on the free Colab T4 GPU available to us; we therefore did not use it, and list it as future work.

4. Experiments

All models were trained on the same leak-free split in Google Colab (NVIDIA T4, PyTorch). The Assignment 1 baselines were re-implemented and re-trained on this split for a fair comparison: HOG + RBF-SVM (C and γ tuned on validation), the three-block CNN trained from scratch, and MobileNetV2 (ImageNet weights, head training followed by fine-tuning of the last blocks). The final ViT variant was chosen on **validation** macro-F1; the test set was used only once for reporting.

Table 2. Test-set results on the clean split (172 images).

Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)
HOG + SVM (A1 baseline)	0.715	0.774	0.646	0.659
Simple CNN (A1 baseline)	0.116	0.078	0.139	0.099
MobileNetV2 (A1 improved model)	0.547	0.396	0.464	0.427
ViT-B/16, no augmentation	0.878	0.882	0.902	0.877
ViT-B/16, with augmentation (selected)	**0.884**	**0.892**	**0.899**	**0.891**

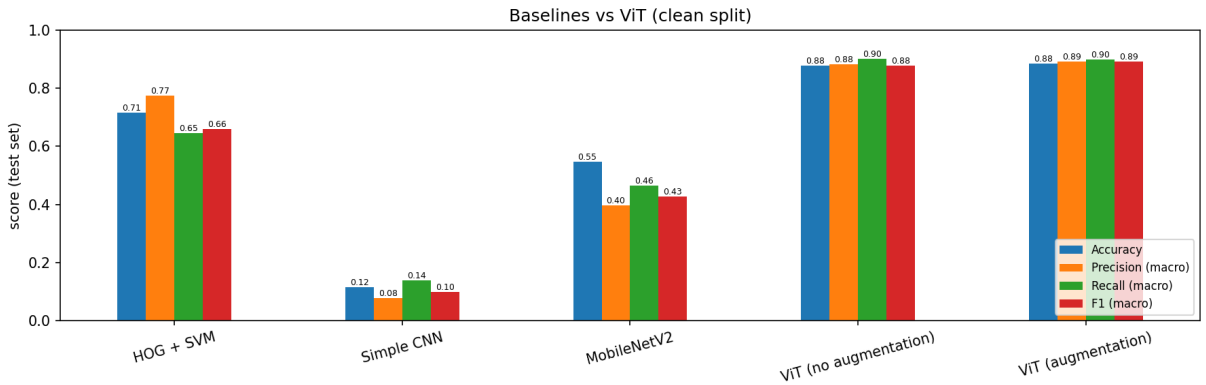


Figure 3. Baselines vs. ViT on the clean split.

Baseline vs. SOTA. The ViT improves macro-F1 by +0.23 over the strongest baseline (HOG + SVM) and by +0.46 over MobileNetV2. For reference, Assignment 1 reported 0.889 accuracy for HOG + SVM and 0.600 for MobileNetV2; those values came from the leaky per-image split and are not comparable — the drop to 0.715 and 0.547 shows how much the leak had inflated them. HOG + SVM recognises every neutral face but misses many happy and sad ones, and MobileNetV2 never predicts *sad*, confusing it with happy (Figure 4). The scratch CNN performs below chance (0.116 accuracy): its validation loss rose from the first epoch, so early stopping kept almost untrained weights, and the network predicts no neutral faces at all. Our hypothesis is that 806 training images from only 66 photo groups are far too few to learn visual features from scratch.

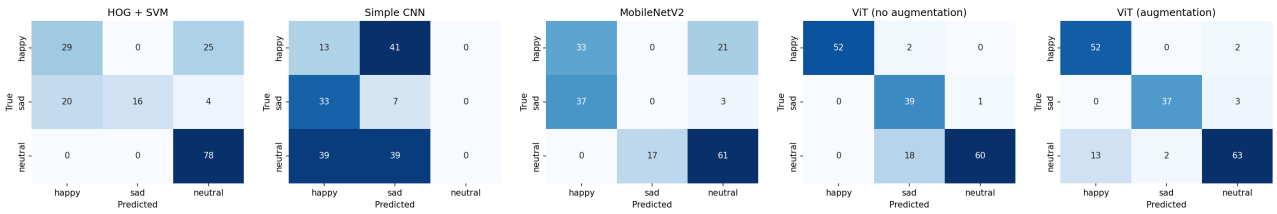


Figure 4. Test confusion matrices (rows: true class; columns: predicted class) — from left: HOG + SVM, Simple CNN, MobileNetV2, ViT without augmentation, ViT with augmentation.

Table 3. Ablation — the only difference is on-the-fly augmentation during ViT training.

Class	ViT, no augmentation (P / R / F1)	ViT, with augmentation (P / R / F1)
Happy	1.000 / 0.963 / 0.981	0.800 / 0.963 / 0.874
Sad	0.661 / 0.975 / 0.788	0.949 / 0.925 / 0.937
Neutral	0.984 / 0.769 / 0.863	0.926 / 0.808 / 0.863
Macro-F1	0.877	0.891

Ablation. Augmentation raised macro-F1 slightly (0.877 → 0.891) and changed the error pattern: without it, 18 neutral faces were called *sad*; with it, sad precision rose from 0.66 to 0.95 and the remaining errors are 13 neutral faces called *happy*. A difference of 0.014 corresponds to only a few images, so we read it as "augmentation does not hurt and makes sad predictions more reliable" rather than as a clear win. The training curves (Figure 5) show that the ViT reaches 100 % training accuracy after one epoch while validation loss rises — strong overfitting, controlled by early stopping. Validation accuracy (0.56–0.67) is clearly lower than test accuracy (0.88): both sets contain only 41–42 independent photo groups, so which groups fall into which set strongly affects the score. In earlier runs with different splits we saw ViT scores vary by several points, so the exact numbers should be read with this uncertainty in mind; the ranking of the models, however, was the same.

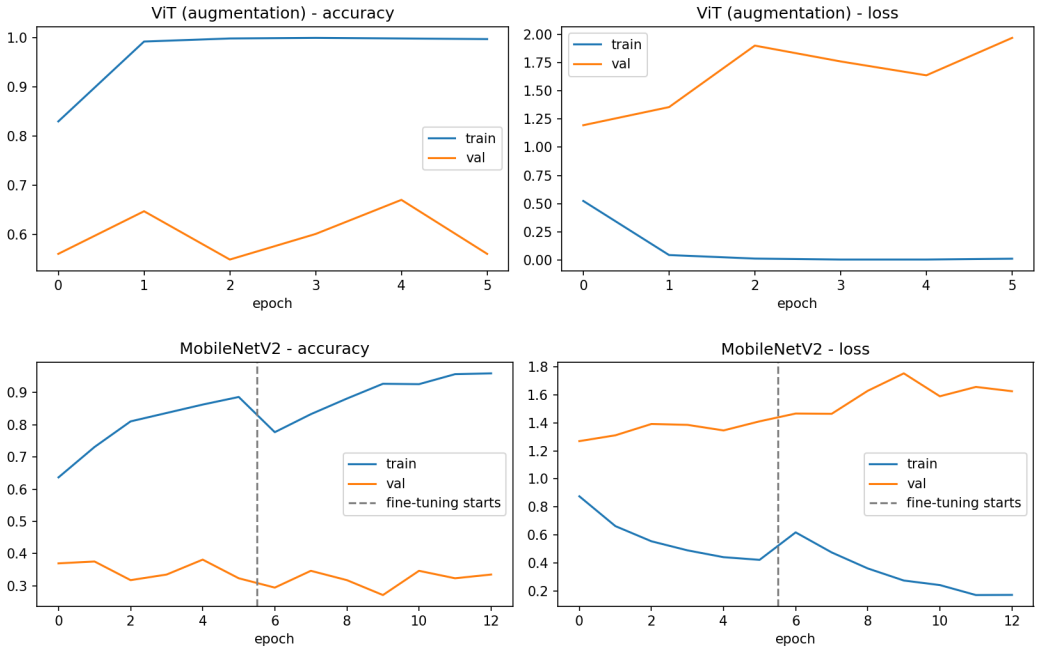


Figure 5. Training curves of the selected ViT (top) and of MobileNetV2 (bottom; the dashed line marks the start of fine-tuning).

Table 4. Inference latency (batch 1, mean of 30 runs, Google Colab).

Model	Parameters	CPU (ms / image)	GPU T4 (ms / image)
Simple CNN	0.10 M	28.6	1.4
MobileNetV2	2.23 M	24.2	5.7
ViT-B/16	85.8 M	474.5	18.9

Accuracy vs. latency. The ViT is 38 times larger than MobileNetV2 and about 20 times slower on CPU (0.47 s per image), but at 19 ms per image on a GPU it still runs at more than 50 frames per second. For a CPU-only or mobile deployment, a smaller transformer (e.g. ViT-Small or a distilled model) would be the natural next step.

5. Evaluation under Animation

Table 5. ViT on the 24 source photos (stills) vs. their animated frames. Consistency = share of frames that agree with the clip's majority prediction.

Animation	Clips	Frames	Macro-F1 stills	Macro-F1 frames	Δmacro-F1	Consistency	Majority = true label
pose (head motion)	24	624	0.870	0.880	**+0.010**	0.963	87.5 %
lip (talking mouth)	24	624	0.870	0.483	**−0.388**	0.772	33.3 %

With head motion only, the ViT is robust: macro-F1 is unchanged and 96 % of frames agree with the clip's majority prediction. Talking is a different story. In *lip* mode, frame accuracy stays at 1.00 for happy clips but drops to 0.34 for neutral and 0.23 for sad clips, because an open, moving mouth is read as a smile: in Figure 6 (right), a neutral face flips to *happy* with 0.85–0.88 probability while the mouth moves and returns to *neutral* when it closes. This agrees with the attention maps, which concentrate on the mouth. Part of the drop may be genuine, because the mouth shapes copied from the driving video can resemble smiling, so the source photo's label is not guaranteed to remain correct for every frame. Practically, a video-based recogniser should smooth predictions over time and be trained with talking faces.

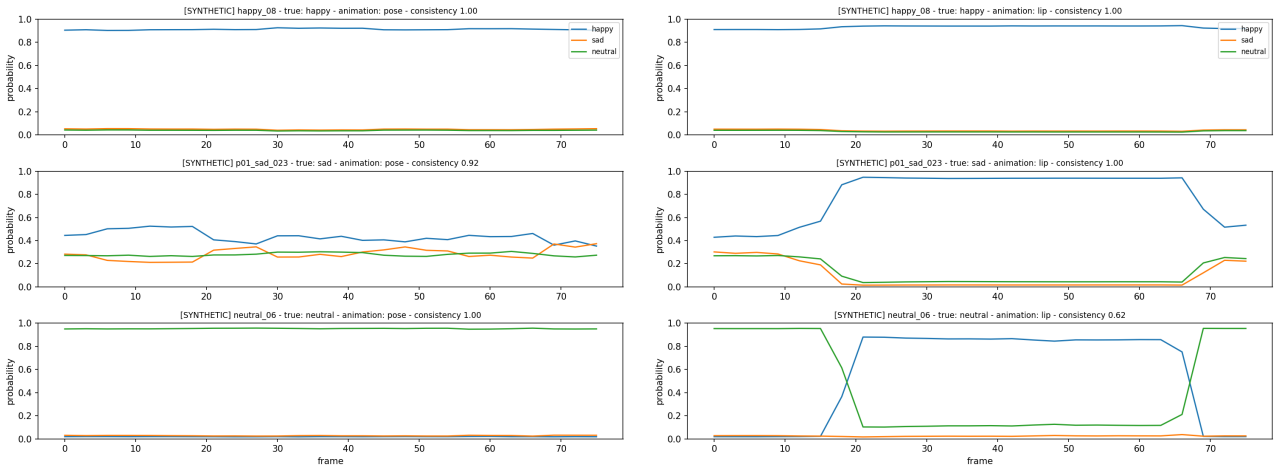


Figure 6. Emotion timelines of three SYNTHETIC clips (one per emotion): left, *pose* animation; right, *lip* animation of the same photos.

6. Explainability

We visualise the ViT with attention rollout [2]: the head-averaged attention matrices of all 12 layers are multiplied, with the identity added for the residual connections, and the attention of the [CLS] token to the 196 patches is shown as a heatmap. For correctly classified test faces the heat lies on the mouth and lips for happy faces, on the eyes and nose for sad faces, and on the nose and central face for neutral faces (Figure 7), i.e. on the regions that carry expressions rather than on hair or clothing. Some heat also falls on background corners, a known artefact of rollout. The same maps stay on the face across animated frames (computed in the animation notebook).

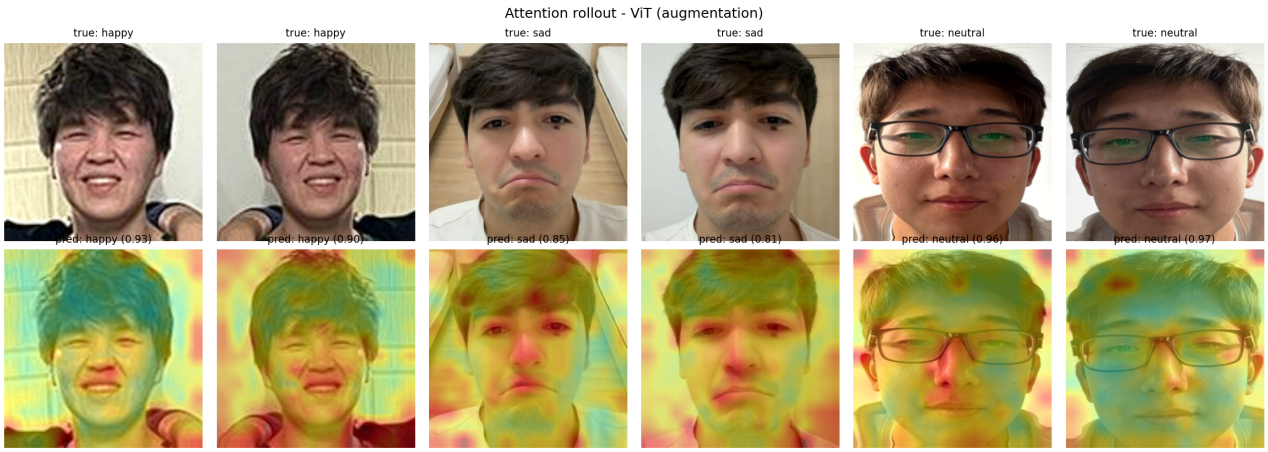


Figure 7. Attention rollout of the selected ViT on test photos (top: input with true label; bottom: heatmap with prediction and confidence).

7. Demo Application

The demo is a Gradio web app: the user uploads a photo or takes one with the webcam, and the app shows the probabilities of the three emotions, the attention-rollout heatmap, and the measured inference time together with the environment (device and library versions). Synthetic talking-head clips are generated offline in Colab, as the assignment recommends for heavy generators, and are shown only with a burned-in "SYNTHETIC — AI-generated" banner. The app states on screen that images are processed only for the prediction, are not stored, and that only photos of people who agreed should be used.

8. Ethics and Limitations

Consent and disclosure. All photographed participants agreed to the use of their photos for this project. Every generated clip is labelled as synthetic, both in the file and in every figure, and only test photos of consenting participants were animated. **Misuse.** Talking-head generation is the technology behind deepfakes; Microsoft withheld VASA-1 for this reason [4]. We used animation only to stress-test our classifier and do not publish the generated clips beyond the course. Emotion recognition itself can be misused for surveillance or for judging people in hiring or education; our model is a course prototype and must not be used for decisions about people. **Bias.** The dataset comes from a small, demographically narrow group of young adult volunteers, so performance on other ages, skin tones, cultures or camera conditions is unknown and probably lower. **Limitations.** (i) Only 149 independent photo groups — the test set has 172 images but only 41 groups, so scores have wide uncertainty and changed between splits; (ii) expressions are posed, and the line between a slight smile and a neutral face is subjective; (iii) a single split was used instead of cross-validation; (iv) the *lip* animation may change the true expression; (v) ArtAug was not evaluated.

9. Reproducibility

Environment: Google Colab, NVIDIA T4 GPU, Python 3, PyTorch, torchvision, Hugging Face Transformers, scikit-learn, scikit-image, OpenCV, Gradio; random seed 42. **Inputs:** the dataset folder `dataset/{happy,sad,neutral}/` in Google Drive. **Steps:** (1) run `A2_ViT_emotion_recognition.ipynb` — audits and splits the data (saving `splits.csv`, `test_sources.csv` and the check tables), trains all models, writes every table and figure in this report, exports the weights to `vit_emotion/` (`model.safetensors`, `config.json`, `app_config.json`) and starts the Gradio demo; (2) run `A2_talking_head_clips.ipynb` — installs LivePortrait and its public weights, animates the test photos and writes the animation tables, timelines and heatmaps. Both notebooks run end to end with "Run all" (about 30 and 20 minutes). The repository, app run guide and demo video links are provided with the Moodle submission.

10. Conclusion

On a cleaned, leak-free version of our dataset, a fine-tuned Vision Transformer clearly outperforms all Assignment 1 baselines (macro-F1 0.891 vs. 0.659 at best) and looks at the expressive regions of the face. The VASA-1-style animation test adds a finding that still images could not reveal: the model is robust to head

motion but confuses talking with smiling. Future work: more participants and person-level splits with cross-validation, training on video frames or talking faces with temporal smoothing, a smaller transformer for CPU deployment, and ArtAug-based enhancement of hard cases on a larger GPU.

References

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